

Day 10 - EDA & Class Separability

Distribution Visualization

Today, I performed Exploratory Data Analysis (EDA) to understand how the received signal values are distributed for the two classes.

I used two plots:

1. Histogram

The histogram showed that the two classes are clearly separated.

- Label 0 values are approximately between 0.335 and 0.366.
- Label 1 values are approximately between 0.518 and 0.625.

There was no overlap between the two groups.

2. Boxplot

The boxplot also showed a clear difference between the two classes.

Label 0 values are closely grouped around approximately 0.345 to 0.35.

Label 1 values have a slightly wider range, mainly around 0.52 to 0.535, with some higher values reaching approximately 0.625.

There is a large gap between approximately 0.37 and 0.51 where no signal values are present. This clearly shows that the two classes are separated.

Class Overlap and Turbidity Condition

At the available turbidity condition of 1–2 NTU, there is no overlap between Label 0 and Label 1.

However, the current dataset contains raw data for only one turbidity condition. Because of this, I am not able to study how the signal behaves at different NTU levels.

This is one limitation of the currently available dataset.

Signal Range Analysis

The dataset contains only one numerical input feature called feature, which represents the received signal value.

Because there is only one feature, creating a feature correlation matrix is not useful.

From the analysis, it appears that the feature value alone can clearly determine whether the label is 0 or 1 for the available turbidity condition.

There were no overlapping or unclear values between the two classes.

Modelling Hypotheses for Week 3

Based on the EDA results, I have the following expectations for Week 3:

1. A fixed threshold method and machine learning models such as Logistic Regression, SVM, and Random Forest are expected to perform very well because the two classes are clearly separated.
2. A preliminary test using a fixed threshold and Logistic Regression already gave an accuracy and F1-score of 1.0.
3. Since the classes are already very easy to separate at the available turbidity condition, this dataset alone may not be enough to show whether machine learning performs better than threshold detection when turbidity increases.
4. To properly compare model performance at different turbidity levels, raw data from higher NTU conditions would be needed.